

INFORMATION THEORY & CODING

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Outline

- *Most* of the basic *definitions* will be introduced.

Entropy, Joint Entropy, Conditional Entropy, Relative Entropy, Mutual Information, etc.

- *Relationships* and Applications

Chain Rules, Jensen's, Data-processing, Fano Inequalities, etc.

What is *Information*?

- To have a **quantitative** measure of information contained in an event, we consider intuitively the following:
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 - Information contained in events should be defined in terms of some measure of the *uncertainty* of the events.
 - **Monotonous:** *Less certain* events should contain *more information* than more certain events.
 - **Additive:** The information of *unrelated/independent* events taken as a single event should *equal the sum* of the information of the unrelated events.

Information Measure

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$$I(A)_{\text{self-info}} = -\log \Pr(A).$$

If $\Pr(A) > \Pr(B)$, then $I(A) < I(B)$.

If A, B are independent, then $I(A + B) = I(A) + I(B)$.

Information Measure: Why Logarithm?

1. It is **practically more useful**. Parameters, e.g., time, bandwidth, number of relays, etc., tend to vary **linearly** with the **logarithm** of $\#$ of possibilities.

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3. It is **mathematically more suitable**.

Information Unit

$\log_2$: bit
 $\log_e$: nat
 $\log_{10}$: Hartley (dit)

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$$\log_a X = \frac{\log_b X}{\log_b a} = \log_a b \cdot \log_b X$$

Information (Source)

$S_1, S_2, \dots, S_q$: *Source alphabet*

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The *average* information of the Source is

$$I(\mathcal{S}) = \sum_{i=1}^q p_i \log\left(\frac{1}{p_i}\right),$$

where $\log \frac{1}{p_i}$ is called *self-information* of X .

Entropy

- **Definition.** The *entropy* of a discrete random variable X is given by

$$\begin{aligned} H(X) &= \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{p(x)} \\ &= - \sum_{x \in \mathcal{X}} p(x) \log p(x) \\ &= E \log \frac{1}{p(x)}. \end{aligned}$$

By convention, let $0 \log 0 = 0$ since $x \log x \rightarrow 0$ as $x \rightarrow 0$.

Entropy

■ **Lemma 2.1.1** $H(X) \geq 0$.

Proof. Since $0 \leq p(x) \leq 1$, we have $\log \frac{1}{p(x)} \geq 0$.

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- **Lemma 2.1.2** $H_b(X) = (\log_b a) H_a(X)$.

Proof. Since $\log_b p = \log_b a \log_a p$.

Entropy: An Example

■ Example 2.1.1 Let

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$$X = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - p. \end{cases}$$

$$H(X) = -p \log p - (1 - p) \log(1 - p) = H(p).$$

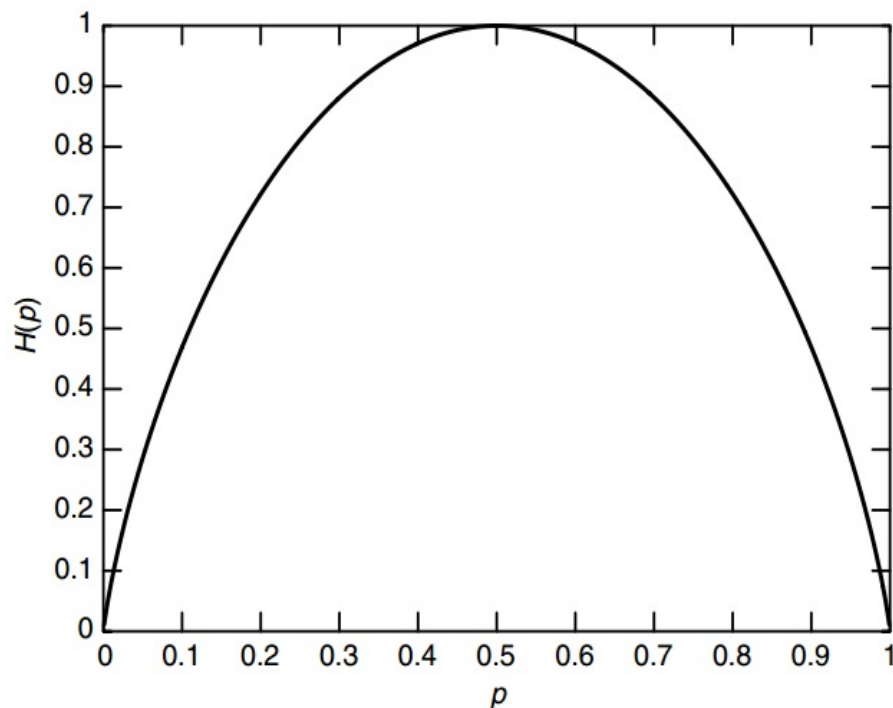


FIGURE 2.1. $H(p)$ vs. p .

Joint Entropy

- **Definition.** The *joint entropy* $H(X, Y)$ of a pair of discrete random variables (X, Y) with a joint distribution $p(x, y)$ is

$$\begin{aligned} H(X, Y) &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y) \\ &= -E \log p(X, Y). \end{aligned}$$

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If X and Y are *independent*, then

$$\begin{aligned} H(X, Y) &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x)p(y) \log p(x)p(y) \\ &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x)p(y) \log p(x) - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x)p(y) \log p(y) \\ &= \sum_{y \in \mathcal{Y}} p(y)H(X) + \sum_{x \in \mathcal{X}} p(x)H(Y) \\ &= H(X) + H(Y). \end{aligned}$$

Conditional Entropy

- If $(X, Y) \sim p(x, y)$, the *conditional entropy* $H(Y|X)$ is

$$\begin{aligned} H(Y|X) &= \sum_{x \in \mathcal{X}} p(x) H(Y|X = x) \\ &= - \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x) \\ &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x) \\ &= -E \log p(Y|X). \end{aligned}$$

Chain Rule

- **Theorem 2.2.1** (*Chain Rule*)

$$H(X, Y) = H(X) + H(Y|X).$$

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$$\log p(X, Y) = \log p(X) + \log p(Y|X)$$

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- **Corollary**

$$H(X, Y|Z) = H(X|Z) + H(Y|X, Z).$$

Chain Rule

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$$H(X, Y) = H(X) + H(Y|X).$$

- **Corollary**

$$H(X, Y|Z) = H(X|Z) + H(Y|X, Z).$$

Proof ?

Example

- **Example 2.2.1** Let (X, Y) have the following *joint distribution*:

$Y \backslash X$	1	2	3	4
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0

What are $H(X)$, $H(Y)$, $H(X, Y)$, $H(X|Y)$, and $H(Y|X)$?

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What are $H(X)$, $H(Y)$, $H(X, Y)$, $H(X|Y)$, and $H(Y|X)$?

$$H(X) = \frac{7}{4} \text{ bits}, H(Y) = 2 \text{ bits}, H(X|Y) = \frac{11}{8} \text{ bits}, \\ H(Y|X) = \frac{13}{8} \text{ bits}, H(X, Y) = \frac{27}{8} \text{ bits}$$

Relative Entropy

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- The *entropy* of a random variable is a measure of *the amount of information* required to describe the random variable.
- The *relative entropy* $D(p||q)$ is a measure of *the distance between two distributions*. We need $H(p)$ bits on average to describe a random variable with distribution p , and need $H(p) + D(p||q)$ bits on average to describe this random variable if we use the code for a distribution q .

Relative Entropy

- The *relative entropy* or *Kullback-Leibler distance* between two probability mass functions $p(x)$ and $q(x)$ is defined as

$$\begin{aligned} D(p||q) &= \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)} \\ &= E_p \log \frac{p(X)}{q(X)}. \end{aligned}$$

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By convention, $0 \log \frac{0}{0} = 0$, $0 \log \frac{0}{q} = 0$ and $p \log \frac{p}{0} = \infty$.

Relative Entropy: Example

Example 2: Consider a horse race with 8 horses. It was determined that the probability of horse i winning is

$$\Pr[\text{horse } i \text{ wins}] = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}\right)$$

Answer 1: Let's try the code of the previous example.

Outcome	Probability	Representation 1
0	$\frac{1}{2}$	000
1	$\frac{1}{4}$	001
2	$\frac{1}{8}$	010
3	$\frac{1}{16}$	011
4	$\frac{1}{64}$	100
5	$\frac{1}{64}$	101
6	$\frac{1}{64}$	110
7	$\frac{1}{64}$	111

To represent a given outcome, the average number of bits is $\bar{\ell} = 3$.

Relative Entropy: Example

Answer 2: What if we allow the length of each representation to vary amongst the outcomes, e.g., a Huffman code:

Outcome	Probability	Representation 2
0	$\frac{1}{2}$	0
1	$\frac{1}{4}$	10
2	$\frac{1}{8}$	110
3	$\frac{1}{16}$	1110
4	$\frac{1}{64}$	111100
5	$\frac{1}{64}$	111101
6	$\frac{1}{64}$	111110
7	$\frac{1}{64}$	111111

The **average** number of bits is

$$\bar{\ell} = \frac{1}{2} + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{16} \cdot 4 + \frac{4}{64} \cdot 6$$
$$= 2$$

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What are $D(p||q)$ and $D(q||p)$?

Relative Entropy

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Example 2.3.1 Let $\mathcal{X} = \{0, 1\}$ and consider two distributions p and q on $\mathcal{X}$. Let $p(0) = 1 - r$, $p(1) = r$, and let $q(0) = 1 - s$, $q(1) = s$. Then

$$\begin{aligned} D(p||q) &= (1 - r) \log \frac{1 - r}{1 - s} + r \log \frac{r}{s} \\ D(q||p) &= (1 - s) \log \frac{1 - s}{1 - r} + s \log \frac{s}{r}. \end{aligned}$$

Relative Entropy

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$$D(q||p) = (1 - s) \log \frac{1 - s}{1 - r} + s \log \frac{s}{r}.$$

In general, $D(p||q) \neq D(q||p)$!

Mutual Information

- **Definition.** Consider two random variables X and Y with a joint probability mass function $P(x, y)$ and marginal probability mass functions $p(x)$ and $p(y)$. The *mutual information* $I(X; Y)$ is the relative entropy between the joint distribution and the product distribution $p(x)p(y)$:

$$\begin{aligned} I(X; Y) &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \\ &= D(p(x, y) || p(x)p(y)) \\ &= E_{p(x, y)} \log \frac{p(X, Y)}{p(X)p(Y)}. \end{aligned}$$

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The *mutual information* is a measure of the amount of information that one random variable contains about another random variable.

■ **Theorem 2.4.1** (*Mutual information and entropy*)

$$I(X; Y) = H(X) - H(X|Y)$$

$$I(X; Y) = H(Y) - H(Y|X)$$

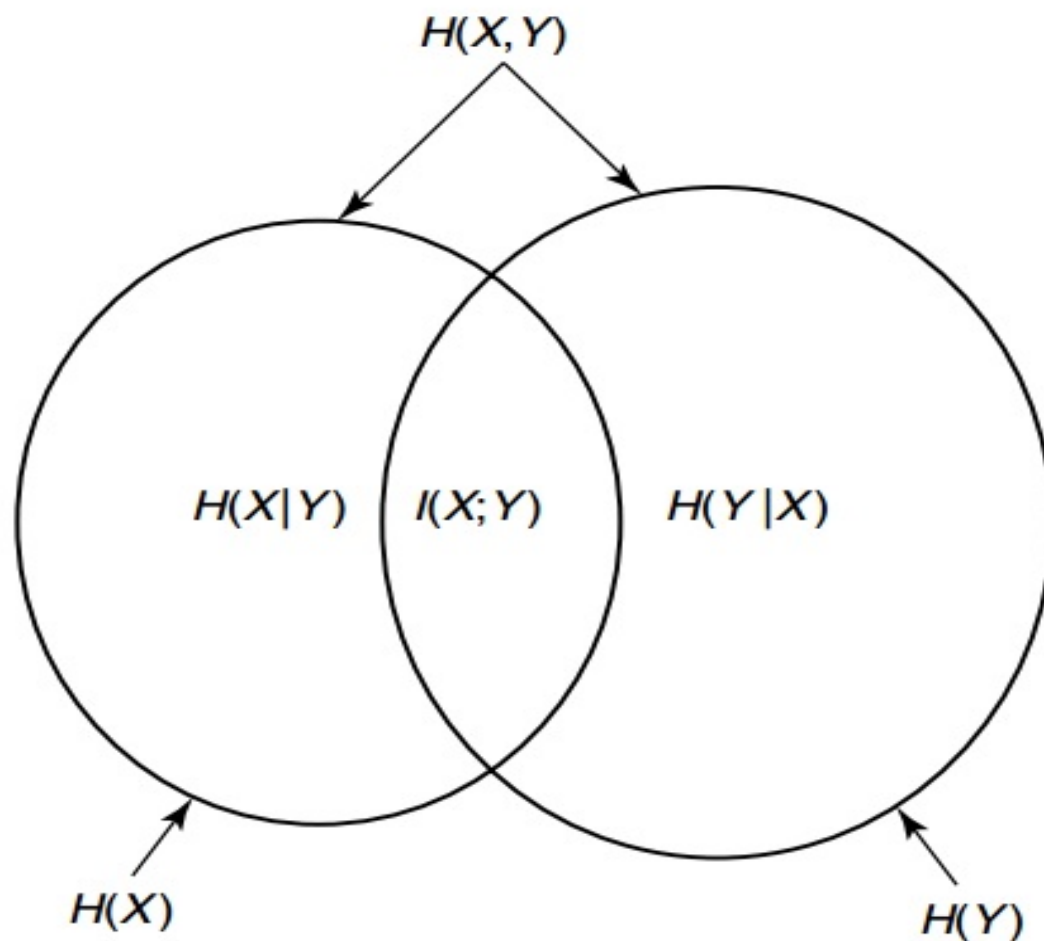
$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = I(Y; X)$$

$$I(X; X) = H(X).$$

Relationships

- *Mutual information and entropy*



Chain Rules

- **Theorem 2.5.1** (*Chain rule for entropy*)
Let $X_1, X_2, \dots, X_n$ be drawn according to $p(x_1, x_2, \dots, x_n)$. Then

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1).$$

Proof.

Chain Rules for Information

- **Definition** The *conditional mutual information* of random variables X and Y given Z is defined by

$$\begin{aligned} I(X; Y|Z) &= H(X|Z) - H(X|Y, Z) \\ &= E_{p(x,y,z)} \log \frac{p(X, Y|Z)}{p(X|Z)p(Y|Z)}. \end{aligned}$$

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- **Theorem 2.5.2** (*Chain rule for mutual information*)

$$I(X_1, X_2, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y|X_{i-1}, X_{i-2}, \dots, X_1).$$

Conditional Relative Entropy

- **Definition.** For joint probability mass functions $p(x, y)$ and $q(x, y)$, the *conditional relative entropy* $D(p(y|x)||q(y|x))$ is the average of the relative entropies between the conditional probability mass functions $p(y|x)$ and $q(y|x)$ averaged over the probability mass function $p(x)$. More precisely,

$$\begin{aligned} D(p(y|x)||q(y|x)) &= \sum_x p(x) \sum_y p(y|x) \log \frac{p(y|x)}{q(y|x)} \\ &= E_{p(x,y)} \log \frac{p(Y|X)}{q(Y|X)}. \end{aligned}$$

Chain Rules for Relative Entropy

- **Theorem 2.5.3.** (*Chain rule for relative entropy*)

$$D(p(x, y) || q(x, y)) = D(p(x) || q(x)) + D(p(y|x) || q(y|x)).$$

Chain Rules for Relative Entropy

■ **Theorem 2.5.3.** (*Chain rule for relative entropy*)

$$D(p(x, y) || q(x, y)) = D(p(x) || q(x)) + D(p(y|x) || q(y|x)).$$

Proof.

$$\begin{aligned} D(p(x, y) || q(x, y)) &= \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{q(x, y)} \\ &= \sum_x \sum_y p(x, y) \log \frac{p(x)p(y|x)}{q(x)q(y|x)} \\ &= \sum_x \sum_y p(x, y) \log \frac{p(x)}{q(x)} + \sum_x \sum_y p(x, y) \log \frac{p(y|x)}{q(y|x)} \\ &= D(p(x) || q(x)) + D(p(y|x) || q(y|x)). \end{aligned}$$

Next Week...

Inequalities and their applications.

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